ERDŐS INSTITUTE – SUMMER 2026

Can Market Signals Improve Delta Hedging?

A leakage-controlled replication experiment on historical SPY paths

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Quantitative Finance Project

We test hedge inputs – not historical option execution

Can point-in-time signals reduce terminal replication error versus fixed-volatility Black–Scholes?

Controlled comparison
Same payoff, initial premium, hedge calendar, and transaction-cost rule.

OBSERVED MARKET DATA

- SPY adjusted prices and volume
- VIX, XLK returns, and short rates

MODEL-DEFINED CLAIMS

- European calls with controlled maturity and moneyness
- No historical option bids, asks, or contract IV

Four strategies; one common delta engine

FV-BS

Fixed 21-day realized volatility

Benchmark fixed at contract initiation.

ROLLING-BS

Updated 21-day realized volatility

Uses the latest price information.

VIX-BS

Daily VIX as hedge volatility

Adds forward-looking option-market information.

MSA-DELTA

Rolling vol $\times$ risk multiplier

Combines seven market signals.

Common premium · daily rebalancing · identical payoff · identical costs

Definition: volatility changes delta away from the money

COMMON HEDGE FORMULA

$$\Delta_t = N(d_{1,t}), \qquad d_{1,t} = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma_t^2)\tau}{\sigma_t\sqrt{\tau}}.$$

Example contract

$S = 100$, $K = 105$, $\tau = 21/252$, and $r = 3\%$.

NORMAL STATE

$\sigma = 20\% \Rightarrow \Delta = 0.220$

$\downarrow$ $+0.077$ shares per option

STRESS STATE

$\sigma = 28\% \Rightarrow \Delta = 0.297$

Why this matters: volatility adjustments materially change OTM and ITM deltas, but have little effect on an ATM delta near 0.5.

Definition: seven trailing signals create three risk states

STANDARDIZED RISK SCORE

$$R_t = 0.30\,\mathrm{IVRank}_t + 0.20\,\mathrm{VIX}_t + 0.20\,\frac{\mathrm{RV}_{21,t}}{\mathrm{RV}_{63,t}} + 0.10\,\mathrm{VolumeShock}_t + 0.10\,\mathrm{Drawdown}_{63,t} + 0.05\,\mathrm{SectorDiv}_t + 0.05\,|\Delta\mathrm{Rate}_{21,t}|.$$

Features are standardized using development-period statistics only.

NORMAL $R_t < q_{70}$ $m = 1.00$

ELEVATED $q_{70} \le R_t < q_{90}$ $m = 1.15$

STRESS $R_t \ge q_{90}$ $m = 1.40$

$$\sigma_t^{\mathrm{MSA}} = m(Z_t)\widehat{\sigma}_{21,t}$$

The test period stays untouched until final evaluation

TRAIN

2016–2020

Scale features and estimate score thresholds.

VALIDATE

2020–2022

Choose one of three prede-clared MSA settings.

TEST ONCE

2022–2024

Evaluate 1,149 contracts per strategy.

3

maturities: 10, 21, 42 days

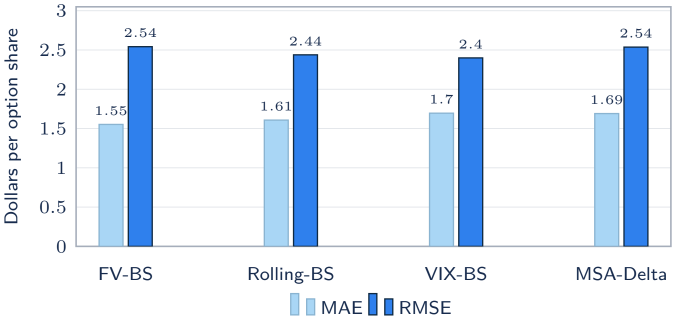
3

strike/spot: 0.95, 1.00, 1.05

5

trading days between starts

No single hedge wins every error metric



1.552

Lowest MAE: FV-BS

2.399

Lowest RMSE: VIX-BS

Interpretation
Average accuracy favors simplicity; large-error control favors forward-looking volatility.

VIX helps most where losses are largest

5.52

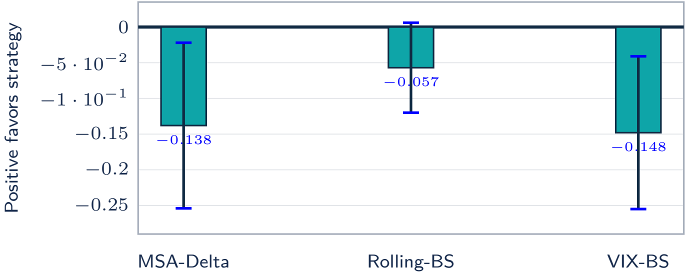
VIX-BS 95% loss Expected Shortfall

6.75

FV-BS 95% loss Expected Shortfall

Lower is better. VIX reduces expected loss in the worst 5% of outcomes by about **18%**.

PAIRED MAE IMPROVEMENT OVER FV-BS



95% moving date-block bootstrap intervals

The ranking survives transaction-cost changes

Transaction cost	Lowest MAE		Lowest RMSE	
0 bps	FV-BS	1.456	VIX-BS	2.394
5 bps	FV-BS	1.552	VIX-BS	2.399
10 bps	FV-BS	1.745	VIX-BS	2.510

STABLE FINDING 1

Fixed volatility remains best for average absolute error.

STABLE FINDING 2

VIX remains best for squared error at every cost level.

Results are also reported for all nine maturity–moneyness cells.

The useful signal is simpler than the model we designed

Forward-looking volatility improves tail-risk control – but more signals do not guarantee a better hedge.

KEEP

VIX-BS as the strongest tail-risk benchmark.

REJECT

A blanket claim that additional complexity improves hedging.

NEXT

Repeat the design using contract-level option quotes when available.

IMPORTANT LIMITATIONS

- VIX is an index-level 30-day proxy, not contract-level SPY IV.
- Controlled European claims omit historical spreads and early exercise.
- Block bootstrap mitigates, but does not eliminate, overlap dependence.